

# Adaptive Preconditioned Dynamics, Solution Geometry, and Generalization in Neural Networks

A Matched-Seed Study of SGD and Adam Across Parameter, Basin, Function, and Representation Geometry

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## Abstract

Adaptive optimizers are often viewed as tools for faster convergence, but in over-parameterized neural networks they may also change which low-loss solution is selected. We study this question by comparing SGD with momentum and Adam as two preconditioned stochastic dynamics under a matched-seed protocol on Fashion-MNIST. Mathematically, SGD applies scalar spatial preconditioning while Adam applies a time-varying diagonal preconditioner, and a local quadratic analysis shows that this changes the effective error dynamics through  $e_{t+1} \approx (I - P_t H)e_t$ . Empirically, Adam moves farther from initialization, traverses a longer path, and selects more dispersed parameter regions. We then test whether these differences persist under a loss-matched control, correspond to Hessian and perturbation-based flatness, imply basin separation through linear mode connectivity, and survive at the function and representation levels through prediction disagreement, symmetric KL, logit cosine, feature-kernel alignment, and centered linear CKA. The results show that, *at this protocol scale (Fashion-MNIST, two small architectures, five matched seeds, 30 epochs)*, Adam changes geometry strongly in parameter space, mildly at the basin and logit levels, and weakly at the representation level. Thus optimizer-induced implicit bias is real, but its generalization effect is filtered through loss level, architecture, basin connectivity, function-space equivalence, and representation similarity.

## 1 Introduction

Modern neural networks are typically over-parameterized: they can fit the training data in many different ways, often with many parameter settings achieving nearly zero training error. In this regime, the optimizer does more than minimize the empirical risk. It also helps select which interpolating or near-interpolating solution the training procedure reaches. This selection effect is commonly called the *implicit bias* or *implicit regularization* of the optimization algorithm.

Here we ask whether adaptive optimizers change that implicit bias. We focus on the comparison between SGD with momentum and Adam. SGD applies one global learning-rate scale to all coordinates of the stochastic gradient. Adam instead rescales each coordinate using estimates of the first and second moments of recent gradients [1]. This coordinate-wise adaptation is often helpful in early optimization, especially when gradients vary strongly across layers or coordinates. At the same time, it may also change the geometry of the final solution and therefore affect generalization.

The overall project question is:

*Do adaptive optimizers alter the implicit bias and generalization of neural networks?*

We divide this broad question into three parts that frame the original study:

1. **Early-stage convergence.** Does Adam converge faster than SGD at the beginning of training?
2. **Solution geometry.** If Adam is faster, does it reach a different kind of solution, or only the same solution earlier?
3. **Generalization.** Do the observed geometric differences explain differences in test performance or train–test gap?

The extended report (Parts IV–IX) deepens this question by following the optimizer-induced effect through additional layers of evidence: trajectory geometry (path length, update norm), a loss-matched control that decouples geometry from training-loss level, perturbation flatness as an operational check on the Hessian-trace proxy, linear mode connectivity to test for basin separation, and function-space and representation-space comparisons that ask whether parameter differences propagate to the learned predictor and to internal feature geometry. The contributions are accordingly:

1. We formulate SGD and Adam as two preconditioned stochastic dynamics and develop a local quadratic analysis  $e_{t+1} \approx (I - P_t H)e_t$  that explains why a time-varying diagonal preconditioner can change the selected low-loss solution.
2. We use matched seeds (shared initialization, shared minibatch order) to isolate optimizer-induced effects from initialization and batch-order randomness.
3. We characterize parameter-space geometry through distance from initialization, cumulative path length, directness ratio, Hessian curvature proxies, and inter-seed dispersion, and test fairness via a loss-matched control and an update-norm control.
4. We probe operational geometry through perturbation flatness and linear mode connectivity, separating Hessian-based curvature claims from actual robustness and basin connectivity.
5. We compare the final models in function and representation space using prediction disagreement, symmetric KL, logit cosine similarity, feature-kernel alignment, and centered linear CKA, organizing the results in a three-level synthesis (parameter / basin / function–representation).

The answer is not simply that Adam is better or worse than SGD. Instead, the experiments show a conditional pattern. In the MLP, Adam has a clear early optimization advantage and lands in much flatter solutions according to the Hessian proxies we measure ( $5\times$  lower  $\lambda_{\max}$ ,  $6\times$  lower trace). Even so, these geometric differences do not produce a clear generalization advantage: test accuracy and train–test gap differ by less than the observed seed-to-seed variation, which is not fully explained by a naive flat-minima interpretation. In the CNN, Adam is still slightly faster in training loss, but its dominant Hessian eigenvalue is actually higher than SGD’s, while both optimizers reach similar test accuracy and generalization gaps. In other words, adaptive optimization changes the training trajectory and the selected parameter region, but whether this geometric change matters for generalization depends on architecture.

## 1.1 Scope of Claims

The implicit-bias and generalization claims in this report are claims about Adam-vs-SGD trajectory and solution geometry *at the scale and protocol of this study*: small image-classification architectures on Fashion-MNIST under a matched-seed protocol with no explicit regularization, trained for tens of epochs across a handful of seeds. The MLP/SmallCNN architecture pair is a controlled scale-down designed to expose mechanism, not a representative sample of modern deep learning. We do not claim that the same effects, in the same direction or magnitude, transfer to large-scale settings such as ResNets on ImageNet, transformer language models, or training horizons of  $10^4+$  steps. The full experimental protocol is given in Section 3 and a detailed list of caveats appears in Section 15.2.

## 2 Mathematical Background and Problem Description

Let  $\hat{L}(\theta)$  denote the empirical cross-entropy loss and let

$$g_t = \nabla_{\theta} \hat{L}_{B_t}(\theta_t)$$

be the stochastic gradient on minibatch  $B_t$ . Rather than treating SGD and Adam as two unrelated engineering rules, we view both as preconditioned stochastic dynamics,

$$\theta_{t+1} = \theta_t - P_t q_t,$$

where  $q_t$  is a gradient-like direction and  $P_t$  is the effective preconditioner. For SGD with momentum,

$$u_t = \mu u_{t-1} + g_t, \quad \theta_{t+1} = \theta_t - \eta u_t,$$

so momentum smooths gradients over time, while the spatial scaling remains scalar:  $P_t = \eta I$ . Expanding the recursion,

$$u_t = \sum_{k=0}^t \mu^{t-k} g_k,$$

shows that momentum is a temporal averaging mechanism, not a coordinate-wise metric change.

Adam instead computes

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^{\odot 2},$$

with bias-corrected moments

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}.$$

Its update can be written as

$$\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} = \theta_t - \alpha D_t \hat{m}_t,$$

where

$$D_t = \text{diag} \left( \frac{1}{\sqrt{\hat{v}_{t,i}} + \varepsilon} \right)_i.$$

Thus the key mathematical contrast is scalar spatial preconditioning for SGD versus time-varying diagonal preconditioning for Adam.

This distinction matters in local geometry. Near a low-loss point  $\theta^*$ , approximate

$$\widehat{L}(\theta) \approx \widehat{L}(\theta^*) + \frac{1}{2}(\theta - \theta^*)^\top H(\theta - \theta^*).$$

For  $e_t = \theta_t - \theta^*$ , preconditioned gradient dynamics obey

$$e_{t+1} \approx (I - P_t H) e_t.$$

For SGD, this becomes  $e_{t+1} \approx (I - \eta H) e_t$ . If  $H = Q \Lambda Q^\top$  and  $\tilde{e}_t = Q^\top e_t$ , then

$$\tilde{e}_{t+1,i} = (1 - \eta \lambda_i) \tilde{e}_{t,i}.$$

A single scalar learning rate controls all Hessian eigendirections. For Adam, however,

$$e_{t+1} \approx (I - \alpha D_t H) e_t,$$

and  $D_t$  generally does not commute with  $H$ . In the Hessian eigenbasis,

$$\tilde{e}_{t+1} = (I - \alpha Q^\top D_t Q \Lambda) \tilde{e}_t,$$

which can mix eigendirections. Equivalently, Adam responds to an effective preconditioned curvature

$$H_t^{\text{eff}} = D_t^{1/2} H D_t^{1/2}.$$

This is the central reason Adam should not be interpreted merely as a faster clock for SGD.

In an over-parameterized network, the low-loss set

$$\mathcal{S}_\varepsilon = \{\theta : \widehat{L}(\theta) \leq \varepsilon\}$$

contains many parameter vectors. The optimizer acts as a selection map

$$\theta_{\mathcal{A}}^* = A_{\mathcal{A}}(\theta_0, \{B_t\}, \text{hyperparameters}).$$

Our matched-seed protocol compares  $\theta_{\text{SGD}}^*$  and  $\theta_{\text{Adam}}^*$  under identical initialization and minibatch order, so observed differences are more directly attributable to the update rule.

We use Hessian quantities as local geometry diagnostics. The dominant curvature is

$$\lambda_{\max}(H) = \max_{\|u\|_2=1} u^\top H u,$$

and the aggregate curvature is  $\text{tr}(H) = \sum_i \lambda_i(H)$ . For a small perturbation,

$$\Delta \widehat{L}(\delta) \approx \frac{1}{2} \delta^\top H \delta.$$

If  $\delta \sim \mathcal{N}(0, \sigma^2 I)$ , then

$$\mathbb{E}[\Delta \widehat{L}(\delta)] \approx \frac{\sigma^2}{2} \text{tr}(H).$$

This identity motivates the perturbation-flatness experiment later in the report.

Finally, sharpness must be interpreted carefully. Euclidean Hessian spectra are parameter-space diagnostics, not invariant measures of function complexity. For ReLU networks, rescaling one layer and inversely rescaling the next can leave the represented function unchanged while changing parameter norms and curvature. Therefore our Hessian and perturbation metrics are used as controlled within-protocol diagnostics, and we later test whether parameter differences survive in basin, function, and representation space.

## 2.1 Project Hypotheses

We test three linked hypotheses. First, Adam should reduce training loss faster than default SGD because diagonal preconditioning adapts coordinate-wise step scales. Second, Adam and SGD should select different parameter-space regions even under matched initialization and minibatch order. Third, parameter-space geometry alone should not fully determine generalization; architecture, basin connectivity, function-space similarity, and representation similarity may filter or suppress optimizer-induced differences.

## 3 Experimental Setup

All experiments use Fashion-MNIST with the standard train/test split and normalized grayscale inputs. We compare two architectures: a batch-normalized MLP and a SmallCNN with two convolutional blocks followed by a small classifier. The MLP has weaker image-specific inductive bias, so optimizer-induced effects should be easier to observe; the CNN adds convolution, pooling, and weight sharing, allowing us to test whether architecture compresses optimizer differences.

The main experiments use five seeds,  $\{42, 123, 456, 789, 2024\}$ . For each seed, the SGD and Adam runs start from the same initialization and use the same deterministic data-loader shuffle. The main protocol uses no weight decay, dropout, or data augmentation, so that explicit regularization does not obscure optimizer-induced effects.

| Component     | Setting                                                 |
|---------------|---------------------------------------------------------|
| Dataset       | Fashion-MNIST                                           |
| Batch size    | 128                                                     |
| Epochs        | 30                                                      |
| Loss          | Cross-entropy                                           |
| Seeds         | 42, 123, 456, 789, 2024                                 |
| SGD           | learning rate 0.01, momentum 0.9, weight decay 0        |
| Adam          | learning rate 0.001, betas (0.9, 0.999), weight decay 0 |
| Architectures | Batch-normalized MLP and SmallCNN                       |

Table 1: Main experimental protocol.

The measured quantities are grouped by the kind of question they answer: early convergence (loss curves and threshold crossing), parameter geometry ( $\|\theta_T - \theta_0\|_2$ , path length, update norm, Hessian proxies, inter-seed dispersion), fairness controls (loss-matched checkpoints and update-norm comparison), operational geometry (perturbation flatness and linear mode connectivity), function-space similarity (prediction disagreement, symmetric KL, logit cosine), representation similarity (linear CKA), and generalization (test accuracy and train–test gap).

## 4 Part I: Early-Stage Convergence

Adam’s most immediate expected advantage is faster early training. We therefore record the first global step at which each run crosses fixed training-loss thresholds. This section is mainly motivational: the deeper question is not whether Adam reaches low loss earlier, but whether it selects a different solution once low loss is reached.

| Model    | Threshold | SGD steps | SGD std | Adam steps | Adam std |
|----------|-----------|-----------|---------|------------|----------|
| MLP      | 2.0       | 5.4       | 0.5     | 2.6        | 0.5      |
| MLP      | 1.0       | 12.8      | 1.9     | 6.8        | 0.8      |
| MLP      | 0.5       | 54.2      | 1.3     | 30.8       | 4.6      |
| MLP      | 0.2       | 643.2     | 137.7   | 477.0      | 96.7     |
| SmallCNN | 2.0       | 4.4       | 0.5     | 3.8        | 0.8      |
| SmallCNN | 1.0       | 9.4       | 2.1     | 8.8        | 1.5      |
| SmallCNN | 0.5       | 36.8      | 9.4     | 32.2       | 5.1      |
| SmallCNN | 0.2       | 346.8     | 38.9    | 312.2      | 49.6     |

Table 2: Mean global steps to first reach each training-loss threshold across five seeds. Lower is faster.

Adam reaches every threshold earlier on average. The effect is strongest in the MLP: for the strict threshold 0.2, Adam requires 477.0 steps while SGD requires 643.2 steps. In the CNN, Adam is still faster, but the gap is smaller because both optimizers fit the architecture well. A small learning-rate check on the MLP shows that tuned SGD can narrow parts of this early gap, so we do not claim Adam is universally faster. The defensible conclusion is narrower: Adam is faster than the default SGD baseline under the matched protocol, especially in the MLP. This motivates the next question: does Adam merely arrive earlier, or does it arrive at a different kind of solution?

## 5 Part II: Solution Geometry and Implicit Bias

### 5.1 Question

The second question asks whether Adam and SGD converge to geometrically different solutions, or whether Adam simply arrives at the same kind of solution faster. We use three main endpoint measurements: (1) the L2 distance each optimizer travels from the initialization point, (2) Hessian-based curvature proxies at the final iterate, and (3) the dispersion of final parameter vectors across random seeds. Part IV then adds stronger trajectory diagnostics through path length, directness ratio, and actual update norms.

## 5.2 Distance from Initialization

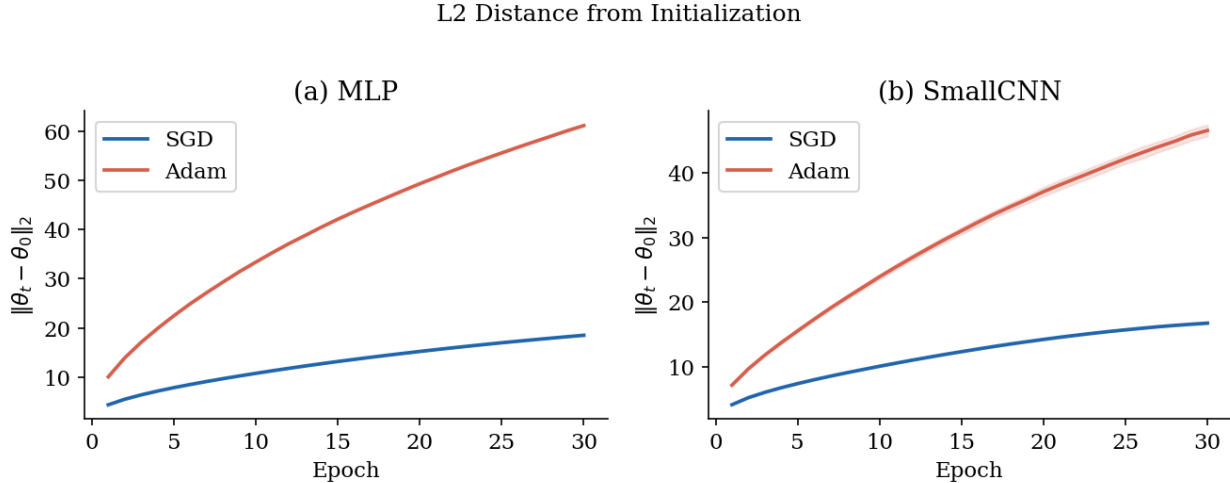


Figure 1: L2 distance from initialization  $\|\theta_t - \theta_0\|_2$  over training epochs. Shaded bands show  $\pm 1$  standard deviation across five matched seeds. Adam moves substantially farther from the starting point in both architectures, and its trajectory is concave (fast early, decelerating later) while SGD drifts linearly.

Figure 1 shows how far each optimizer moves from the shared initialization  $\theta_0$ . In the MLP, Adam reaches a final parameter distance of  $61.20 \pm 0.15$ , while SGD reaches only  $18.49 \pm 0.03$ , a factor of  $3.3\times$ . In the SmallCNN, Adam reaches  $46.79 \pm 0.94$  while SGD reaches  $16.75 \pm 0.09$ , a factor of  $2.8\times$ . Both differences are well outside seed-to-seed variation.

The shapes of the trajectories are qualitatively distinct. Adam’s curve is concave: it advances rapidly in the first five epochs and then decelerates, while SGD follows a more linear drift throughout. This concave trajectory is consistent with Adam’s adaptive preconditioning,

$$\eta_{t,i}^{\text{Adam}} = \frac{\alpha}{\sqrt{\hat{v}_{t,i}} + \varepsilon},$$

and with the changing gradient statistics during training. We should not interpret the curve as evidence for one single mechanism, but it does show that Adam and SGD are not simply following the same path at different speeds. Since both optimizers share identical  $\theta_0$  and data ordering, the large distance difference is most naturally explained by the optimizer update rule.

At the same time, this is a parameter-space statement rather than a direct function-space statement. Neural networks can represent similar functions with different parameter vectors, especially in the presence of scaling symmetries and BatchNorm. Therefore, distance from initialization shows that the optimizer trajectories and selected parameter regions differ, but it does not by itself prove that the learned input–output functions differ by the same amount.

## 5.3 Trajectory Diagnostic Note

We do not use raw gradient norm as evidence of flatness, because Adam updates preconditioned gradients rather than raw gradients. The more relevant trajectory diagnostics are actual parameter movement: distance from initialization here, followed by path length, directness ratio, and update norm in Part IV.

## 5.4 Sharpness and Hessian Trace

We compute two Hessian-based proxies at the final epoch on a fixed mini-batch of 512 examples in evaluation mode.

**Sharpness  $\lambda_{\max}$  via power iteration.** Starting from a random unit vector  $v_0$ , we iterate

$$v_{k+1} = \frac{H(\hat{\theta}) v_k}{\|H(\hat{\theta}) v_k\|_2}, \quad \lambda_{\max} \approx v_k^\top H(\hat{\theta}) v_k,$$

stopping when  $|\lambda_k - \lambda_{k-1}| < 10^{-4}$  or after 100 iterations. The Hessian-vector product  $H(\hat{\theta})v$  is computed without materializing  $H$  using second-order automatic differentiation (Pearlmutter’s trick):

$$H(\hat{\theta})v = \nabla_\theta \left[ \nabla_\theta \hat{L}(\hat{\theta})^\top v \right].$$

**Hessian trace  $\text{tr}(H)$  via Hutchinson’s estimator.** For  $m = 30$  independent Rademacher vectors  $z^{(i)} \sim \{\pm 1\}^d$ ,

$$\text{tr}(H) = \mathbb{E}[z^\top H z] \approx \frac{1}{m} \sum_{i=1}^m (z^{(i)})^\top H(\hat{\theta}) z^{(i)}.$$

The Hutchinson identity follows from  $\mathbb{E}[zz^\top] = I$  for Rademacher vectors:

$$\mathbb{E}[z^\top H z] = \mathbb{E}[\text{tr}(z^\top H z)] = \mathbb{E}[\text{tr}(H z z^\top)] = \text{tr}(H).$$

The two proxies are complementary:  $\lambda_{\max}$  identifies the single worst direction, while  $\text{tr}(H)$  aggregates curvature across all directions.

| Model    | Opt. | Test acc.           | Gap                 | $\lambda_{\max}(H)$ | $\text{tr}(H)$     | Param dist.      |
|----------|------|---------------------|---------------------|---------------------|--------------------|------------------|
| MLP      | SGD  | $0.8901 \pm 0.0024$ | $0.0904 \pm 0.0027$ | $36.20 \pm 5.97$    | $409.71 \pm 30.44$ | $18.49 \pm 0.03$ |
| MLP      | Adam | $0.8887 \pm 0.0030$ | $0.0912 \pm 0.0026$ | $7.22 \pm 1.34$     | $68.13 \pm 5.04$   | $61.20 \pm 0.15$ |
| SmallCNN | SGD  | $0.9166 \pm 0.0010$ | $0.0779 \pm 0.0028$ | $42.58 \pm 15.71$   | $242.47 \pm 50.04$ | $16.75 \pm 0.09$ |
| SmallCNN | Adam | $0.9150 \pm 0.0009$ | $0.0780 \pm 0.0030$ | $64.17 \pm 13.88$   | $218.91 \pm 61.41$ | $46.79 \pm 0.94$ |

Table 3: Final scalar metrics across five matched seeds (mean  $\pm$  std). Accuracy and gap are reported as fractions. Gap = train accuracy – test accuracy.



Hessian-Based Curvature Proxies at Final Epoch

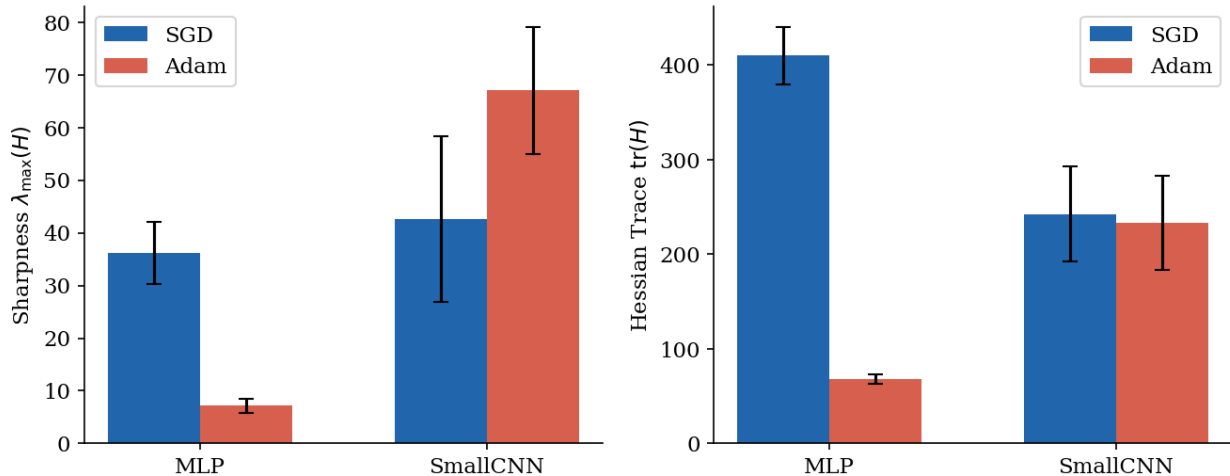


Figure 2: Hessian-based curvature proxies at the final epoch. *Left*: dominant eigenvalue  $\lambda_{\max}(H)$ . *Right*: Hessian trace  $\text{tr}(H)$ . Error bars show  $\pm 1$  std across five seeds. The MLP shows a clear separation: Adam is substantially flatter on both measures. In the SmallCNN, Adam’s  $\lambda_{\max}$  is *higher* than SGD’s, while the traces are comparable.

**MLP.** The difference is large and consistent across all five seeds. SGD reaches solutions with  $\lambda_{\max} = 36.20 \pm 5.97$  and trace  $409.71 \pm 30.44$ , while Adam reaches  $\lambda_{\max} = 7.22 \pm 1.34$  and trace  $68.13 \pm 5.04$ . Adam’s solutions are  $5\times$  flatter in dominant eigenvalue and  $6\times$  lower in total curvature; the error bars of the two optimizers do not overlap in either metric.

The MLP result also clarifies the relationship between distance from initialization and sharpness: Adam travels  $3.3\times$  farther yet ends in a much flatter region. This pattern is plausible geometrically: a long journey can end in a broad valley, while a short journey can end near a narrow basin. In our experiment, the latter pattern describes SGD: it remains close to  $\theta_0$  but arrives in a sharper region.

**SmallCNN.** The CNN result runs in the opposite direction for  $\lambda_{\max}$ : Adam’s mean dominant eigenvalue ( $64.17 \pm 13.88$ ) is higher than SGD’s ( $42.58 \pm 15.71$ ), while the Hessian traces are comparable (218.91 vs. 242.47, a difference smaller than one standard deviation). So the CNN case cannot be summarized as simply “Adam is flatter” or “SGD is flatter.” The architecture’s inductive bias from convolutional weight sharing and pooling appears to constrain both optimizers to regions of comparable overall curvature.

## 5.5 Inter-Seed Parameter Dispersion

We measure the reproducibility of each optimizer’s solutions by computing the mean pairwise L2 distance over all  $\binom{5}{2} = 10$  pairs of final parameter vectors:

$$\bar{d} = \frac{1}{\binom{5}{2}} \sum_{i < j} \|\hat{\theta}_i^* - \hat{\theta}_j^*\|_2.$$

| Architecture | SGD mean pairwise dist. | Adam mean pairwise dist. |
|--------------|-------------------------|--------------------------|
| MLP          | 31.50                   | 87.72                    |
| SmallCNN     | 27.04                   | 67.33                    |

Table 4: Inter-seed parameter dispersion. Adam produces more dispersed final parameter vectors across seeds in both architectures.

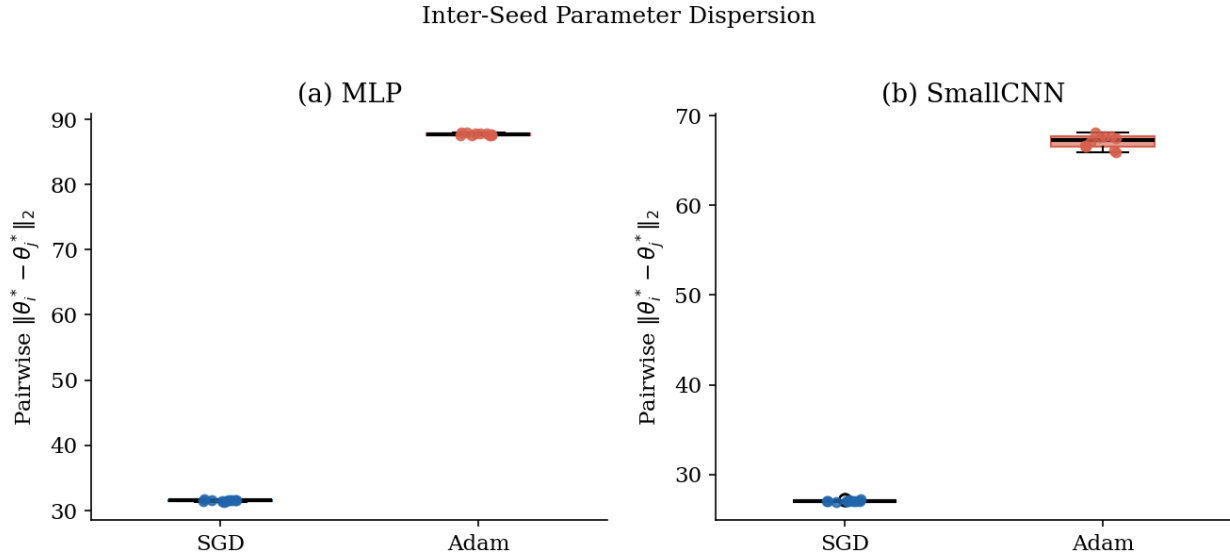


Figure 3: Inter-seed parameter dispersion. Each point is the L2 distance  $\|\hat{\theta}_i^* - \hat{\theta}_j^*\|_2$  between final parameter vectors of two seeds, for a fixed optimizer. SGD points are nearly superimposed in both architectures. Adam solutions are  $2.8\times$  more dispersed in the MLP and  $2.5\times$  more dispersed in the SmallCNN.

Table 4 and Figure 3 show a consistent pattern across both architectures. SGD solutions are tightly clustered across seeds (MLP: 31.50; CNN: 27.04); in Figure 3, the dots are nearly superimposed. Adam solutions are systematically more dispersed: 87.72 for the MLP ( $2.8\times$ ) and 67.33 for the CNN ( $2.5\times$ ). The small spread of the dots within each optimizer means that the pairwise distances are consistent. The main point is that Adam’s pairwise distances are consistently much larger than SGD’s.

This dispersion pattern appears in both architectures tested, including the SmallCNN where curvature evidence is inconclusive. It therefore suggests that the optimizer effect is not limited to the MLP: Adam is more sensitive to the stochastic trajectory and lands in more separated regions across seeds. One plausible explanation is Adam’s coordinate-wise rescaling: across different seeds, small early differences in initialization and minibatch trajectories may affect coordinates differently, and adaptive normalization can allow these differences to accumulate over training.

## 5.6 Part II Conclusion

Together, these four measurements suggest that Adam and SGD select geometrically distinct regions of the loss landscape. In the MLP, Adam travels  $3.3\times$  farther from initialization, takes per-step

updates roughly  $3\times$  larger than SGD’s (Section 8), and converges to solutions that are  $5\times$  flatter in  $\lambda_{\max}$  and  $6\times$  lower in  $\text{tr}(H)$ . Its final solutions are also  $2.8\times$  more dispersed across seeds.

However, the most important finding of Part II is somewhat surprising: despite Adam finding substantially flatter solutions in the MLP, it achieves marginally lower test accuracy (88.87% vs. 89.01%) and a slightly larger train–test gap (9.12% vs. 9.04%) than SGD. These differences are within the observed seed-to-seed variation, so we do not claim a meaningful accuracy or gap advantage for either optimizer. Still, their direction does not support a simple flat-minima explanation. This motivates Part III’s more careful analysis: the link between solution geometry and generalization is not automatic.

In the SmallCNN, Adam still travels farther and produces more dispersed solutions, but the curvature picture reverses: Adam’s  $\lambda_{\max}$  is higher than SGD’s while the traces are comparable, and the generalization metrics remain within seed-to-seed variation.

## 6 Part III: Generalization

We next ask whether the parameter-space differences from Part II explain generalization. The answer is negative at this protocol scale: Adam changes geometry strongly, but neither architecture shows a clear generalization advantage for Adam, even when its Hessian curvature is dramatically lower.

Table 5: Final scalar generalization metrics across five matched seeds (mean  $\pm$  std). Gap = train accuracy – test accuracy.

| Model    | Opt. | Test acc. (%)    | Gap (%)         | $\lambda_{\max}(H)$ |
|----------|------|------------------|-----------------|---------------------|
| MLP      | SGD  | $89.01 \pm 0.24$ | $9.04 \pm 0.27$ | $36.20 \pm 5.97$    |
| MLP      | Adam | $88.87 \pm 0.30$ | $9.12 \pm 0.26$ | $7.22 \pm 1.34$     |
| SmallCNN | SGD  | $91.66 \pm 0.10$ | $7.79 \pm 0.28$ | $42.58 \pm 15.71$   |
| SmallCNN | Adam | $91.50 \pm 0.09$ | $7.80 \pm 0.30$ | $64.17 \pm 13.88$   |

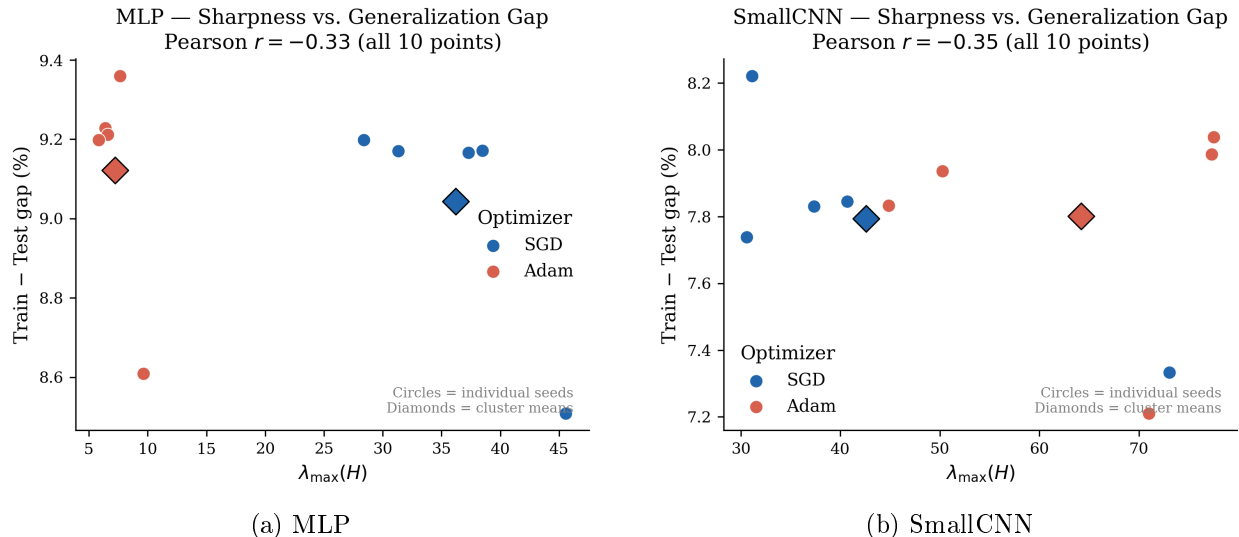


Figure 4: Final-epoch sharpness  $\lambda_{\max}(H)$  versus train-test gap. Each marker is one matched-seed run; diamonds are cluster centroids. In the MLP, the optimizer clusters are well-separated in sharpness ( $\lambda_{\max} \approx 36$  for SGD vs.  $\approx 7$  for Adam) but not in gap. In the SmallCNN, the optimizer clusters overlap on both axes.

**MLP.** Adam reaches solutions with  $\lambda_{\max} = 7.22 \pm 1.34$  versus SGD’s  $36.20 \pm 5.97$  — a  $5\times$  flatness advantage that simple flat-minima accounts [2, 3] would predict to translate into better generalization. The data point in the opposite direction: SGD has marginally higher test accuracy (89.01% vs. 88.87%) and a marginally smaller train-test gap (9.04% vs. 9.12%). Both differences are within seed-to-seed standard deviation, so we do not claim a robust ranking, but the direction is striking: lower Euclidean Hessian sharpness here is not associated with any generalization benefit. Figure 4 visualizes this: the two optimizer clusters are widely separated horizontally (sharpness) but vertically overlapping (gap).

**SmallCNN.** Generalization is essentially tied (test accuracy 91.66% vs. 91.50%; gap 7.79% vs. 7.80%), and the Hessian ordering is mixed (Adam has higher  $\lambda_{\max}$  but comparable trace). The convolutional inductive bias appears to constrain both optimizers to similarly performing solutions even when their parameter trajectories differ.

The combined picture is that the simple flat-minima story is insufficient at this protocol scale. Generalization is mediated by architecture, function-space behavior, representation similarity, and the parameter-space scale-dependence of sharpness diagnostics — themes developed in Parts IV–IX.

## 7 Roadmap for the Extended Analysis

Parts I–III establish the basic pattern: Adam is faster early, selects different parameter-space regions, and yet does not obtain a clear generalization advantage. The remaining sections test this pattern against a sequence of alternative explanations. Part IV asks whether Adam moves farther simply because its effective updates are larger. Part V controls for the possibility that geometry differences are artifacts of comparing endpoints at different training-loss levels. Part VI turns Hessian trace into an operational perturbation test. Part VII asks whether the two endpoints lie in the same low-loss basin or are separated by a barrier. Parts VIII–IX then move beyond parameters

and compare the learned predictors and penultimate-layer representations directly. The goal is not to add unrelated metrics, but to follow one optimizer-induced effect through parameter geometry, basin geometry, function space, representation space, and finally generalization.

## 8 Part IV: Trajectory Geometry: Path Length and Update Norm

Part II established that Adam ends up at a point much farther from  $\theta_0$  than SGD does. By itself this only describes the *net* displacement. Two trajectories can have the same endpoint distance while traversing very different routes. To distinguish “Adam moves farther because it takes larger steps” from “Adam follows a more directed drift,” we now record the *cumulative* parameter movement and the per-step update norm.

### 8.1 Cumulative Path Length

For a training run with  $T$  minibatch updates, define the cumulative path length

$$\mathcal{P}_T = \sum_{t=0}^{T-1} \|\theta_{t+1} - \theta_t\|_2.$$

By the triangle inequality  $\mathcal{P}_T \geq \|\theta_T - \theta_0\|_2$ . The *directness ratio*

$$R_T = \frac{\|\theta_T - \theta_0\|_2}{\mathcal{P}_T} \in (0, 1]$$

quantifies how close the trajectory is to a straight-line drift. A value near 1 means the optimizer moves nearly straight from  $\theta_0$  to  $\theta_T$ ; a small value means most of the parameter motion cancels out.

### 8.2 Update-Norm Profile

The per-step update norm  $\|\Delta\theta_t\|_2 = \|\theta_{t+1} - \theta_t\|_2$  is a stronger signal than the raw gradient norm because it measures the actual change in parameters, including the effect of the optimizer’s preconditioner.

### 8.3 Results

Table 6 summarizes path length, directness ratio, and mean update norm over five matched seeds.

| Model    | Opt. | $\ \theta_T - \theta_0\ _2$ | $\mathcal{P}_T$   | $R_T$               | $\overline{\ \Delta\theta\ }$ |
|----------|------|-----------------------------|-------------------|---------------------|-------------------------------|
| MLP      | SGD  | $18.48 \pm 0.04$            | $464.4 \pm 1.2$   | $0.0398 \pm 0.0001$ | $0.0330 \pm 0.0001$           |
| MLP      | Adam | $61.29 \pm 0.12$            | $1465.8 \pm 1.4$  | $0.0418 \pm 0.0001$ | $0.1042 \pm 0.0001$           |
| SmallCNN | SGD  | $16.79 \pm 0.09$            | $392.0 \pm 3.6$   | $0.0428 \pm 0.0002$ | $0.0279 \pm 0.0003$           |
| SmallCNN | Adam | $46.62 \pm 0.96$            | $1083.8 \pm 21.4$ | $0.0430 \pm 0.0001$ | $0.0770 \pm 0.0015$           |

Table 6: Trajectory geometry: net displacement, cumulative path length  $\mathcal{P}_T$ , directness ratio  $R_T$ , and mean per-step update norm  $\overline{\|\Delta\theta\|}$  across five matched seeds.

The data resolve the path-versus-step question cleanly. In both architectures, Adam’s cumulative path length is roughly 2.8–3.2 $\times$  that of SGD (1465.8 vs. 464.4 on MLP; 1083.8 vs. 392.0 on SmallCNN), and the mean per-step update norm is 2.8–3.2 $\times$  larger (0.1042 vs. 0.0330; 0.0770 vs.

0.0279). The directness ratios are essentially identical for the two optimizers (0.0398 vs. 0.0418 on MLP; 0.0428 vs. 0.0430 on SmallCNN), and both are small in absolute terms ( $\sim 0.04$ ): the trajectories are far from straight, but Adam is no more meandering than SGD relative to its own path length. The factor-three endpoint-distance gap reported in Part II is therefore explained by Adam’s diagonal preconditioner acting as a coordinate-wise step-size amplifier rather than by a qualitatively different geometric strategy. Per-epoch path length curves (Figure 5) and step-norm profiles (Figure 6) confirm that this scale gap is sustained throughout training, not concentrated in a few early epochs.

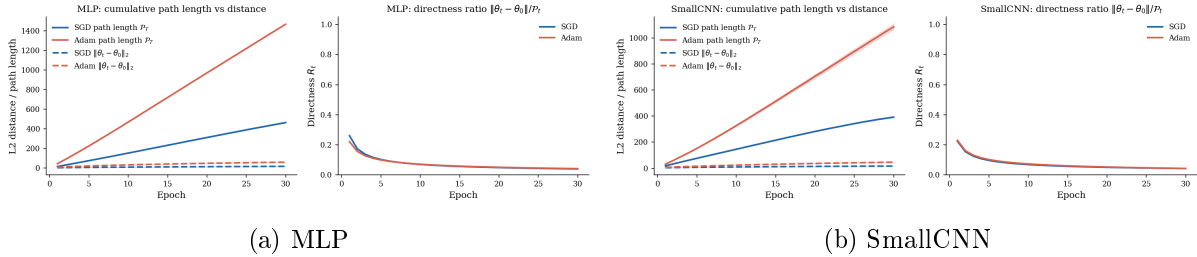


Figure 5: Per-epoch cumulative path length  $\mathcal{P}_t$  versus net displacement  $\|\theta_t - \theta_0\|_2$ . Mean over five matched seeds; bands are  $\pm 1$  std. Adam’s path length grows much faster than its endpoint distance, so the bigger gap is in step size, not direction.

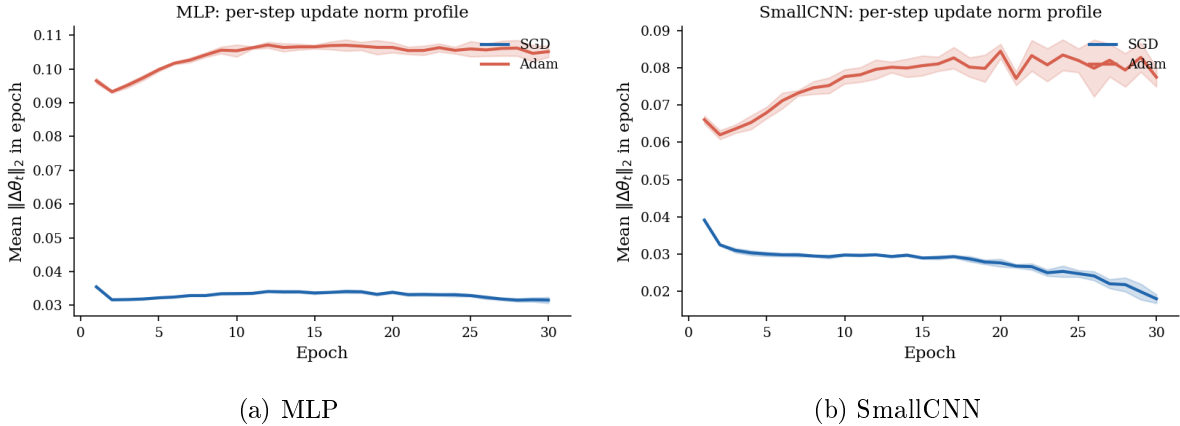


Figure 6: Mean per-step update norm  $\overline{\|\Delta\theta_t\|_2}$  by epoch, five matched seeds.

## 9 Part V: Loss-Matched Geometry Control

### 9.1 Motivation

Fixed-epoch comparisons can confound geometric differences with differences in training-loss level. If at epoch 30 Adam reaches a smaller training loss than SGD does, then the Hessian and distance values being compared are taken at different points along each optimizer’s loss-versus-time curve. A reader could legitimately ask whether the geometric gap is just an artifact of the loss gap. The loss-matched control answers this question.

## 9.2 Method

For each matched seed and architecture, let  $\hat{L}_{\text{Adam}}^{\text{final}}$  be the final training loss of Adam at epoch  $T$ . We then search the SGD trajectory for the epoch

$$t^* = \arg \min_{t \in \{1, \dots, T\}} |\hat{L}_{\text{SGD}}(t) - \hat{L}_{\text{Adam}}^{\text{final}}|,$$

load SGD’s parameter checkpoint at epoch  $t^*$ , and compare its geometry to Adam’s final solution. The compared quantities are the same we used in Part II: distance from initialization, dominant Hessian eigenvalue  $\lambda_{\max}(H)$ , and Hessian trace.

## 9.3 Results

Table 7 reports the loss-matched comparison. The mean matched SGD epoch is  $t^* = 29.8$  for the MLP and  $t^* = 28.8$  for the SmallCNN, indicating that SGD reaches Adam’s final loss late in its 30-epoch trajectory in both architectures.

| Model    | Endpoint          | $\ \theta - \theta_0\ _2$ | $\lambda_{\max}(H)$ | $\text{tr}(H)$   |
|----------|-------------------|---------------------------|---------------------|------------------|
| MLP      | SGD @ epoch $t^*$ | $18.42 \pm 0.10$          | $37.15 \pm 11.65$   | $437.2 \pm 24.3$ |
| MLP      | Adam @ epoch $T$  | $61.29 \pm 0.12$          | $6.86 \pm 0.79$     | $68.4 \pm 9.4$   |
| SmallCNN | SGD @ epoch $t^*$ | $16.58 \pm 0.24$          | $6.17 \pm 12.34$    | $221.8 \pm 64.9$ |
| SmallCNN | Adam @ epoch $T$  | $46.62 \pm 0.96$          | $61.77 \pm 26.43$   | $184.8 \pm 57.0$ |

Table 7: Loss-matched control: SGD parameters at the epoch with closest training loss to Adam’s final loss, paired with Adam’s final parameters. Mean  $\pm$  std across five matched seeds.

**Estimator Variance.** The standard deviations on  $\lambda_{\max}(H)$  in Table 7 are large relative to the means, particularly for the SmallCNN/SGD entry ( $6.17 \pm 12.34$ ). Two sources of noise compound here: power-iteration estimates on a  $\sim 10^5$ -parameter neural Hessian are sensitive to the random initialization of the iterate, and the value of  $\lambda_{\max}$  itself varies across the five matched seeds. We report mean  $\pm$  std for comparability with Part II rather than median/IQR. The MLP architecture comparison ( $37.15$  vs.  $6.86$ ) is statistically distinguishable at one sigma despite this noise, but the SmallCNN reversal ( $6.17$  vs.  $61.77$ ) should be read as a directional finding from a single architecture pair rather than a tight quantitative claim.

The architecture comparison is striking. On the MLP, Adam’s  $\lambda_{\max}$  is roughly  $5.4\times$  smaller than SGD’s at matched training loss ( $6.86$  vs.  $37.15$ ), and its Hessian trace is  $6.4\times$  smaller ( $68.4$  vs.  $437.2$ ): the Part II flatness gap survives the loss-matching control intact. The two optimizers genuinely select different curvature regimes; Adam’s flatter landscape is not a consequence of having simply reached lower training loss. On the SmallCNN, the picture inverts: Adam’s  $\lambda_{\max}$  is roughly  $10\times$  *larger* than SGD’s at matched loss ( $61.77$  vs.  $6.17$ ), while Hessian trace is comparable ( $184.8$  vs.  $221.8$ ). The opposite sign of the curvature gap on the SmallCNN, even after loss matching, suggests that the optimizer-induced flatness ordering is not a universal property of Adam-versus-SGD: at least one of these architecture-optimizer combinations produces a different selection. Given the single architecture pair and the high variance of  $\lambda_{\max}$  estimates noted above, we read this as a directional finding that motivates replication on a larger architecture sample rather than as an established architectural rule.

Function-space disagreement also persists at matched loss. At  $t^*$  versus  $T$ ,  $D_{\text{pred}}$  is  $0.0772 \pm 0.0066$  on the MLP and  $0.0692 \pm 0.0024$  on the SmallCNN, while symmetric KL is  $0.1880$  and  $0.2858$

respectively—values comparable to the standard final-epoch comparison reported in Section 12. Even after controlling for training progress, SGD and Adam have selected different predictors.

## 10 Part VI: Perturbation Flatness

The Hessian-based proxies in Part II are local quadratic measurements. To test whether they reflect actual loss robustness, we evaluate the loss change under explicit parameter perturbations. The local quadratic approximation predicts

$$\widehat{L}(\theta^* + \delta) - \widehat{L}(\theta^*) \approx \frac{1}{2} \delta^\top H(\theta^*) \delta,$$

and for isotropic  $\delta \sim \mathcal{N}(0, \sigma^2 I)$ ,

$$\mathbb{E}_\delta [\widehat{L}(\theta^* + \delta) - \widehat{L}(\theta^*)] \approx \frac{\sigma^2}{2} \text{tr}(H).$$

### 10.1 Relative Global Perturbation

To make perturbation magnitudes comparable across optimizers whose absolute parameter scales differ (Adam typically reaches points with larger  $\|\theta\|_2$ ), we use a *relative* global perturbation:

$$\delta = \sigma \frac{\|\theta^*\|_2}{\|\xi\|_2} \xi, \quad \xi \sim \mathcal{N}(0, I_d).$$

Equivalently,  $\|\delta\|_2 = \sigma \|\theta^*\|_2$ , so  $\sigma$  controls the ratio of perturbation norm to parameter norm. We sweep  $\sigma \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$  and average over  $K = 10$  independent perturbations per  $\sigma$ . Loss is evaluated on a fixed subset of 4096 training examples in evaluation mode (BatchNorm running stats from training).

### 10.2 Results

Figure 7 shows  $\Delta \widehat{L}(\sigma)$  versus  $\sigma$  on log-log axes for both architectures and both optimizers.

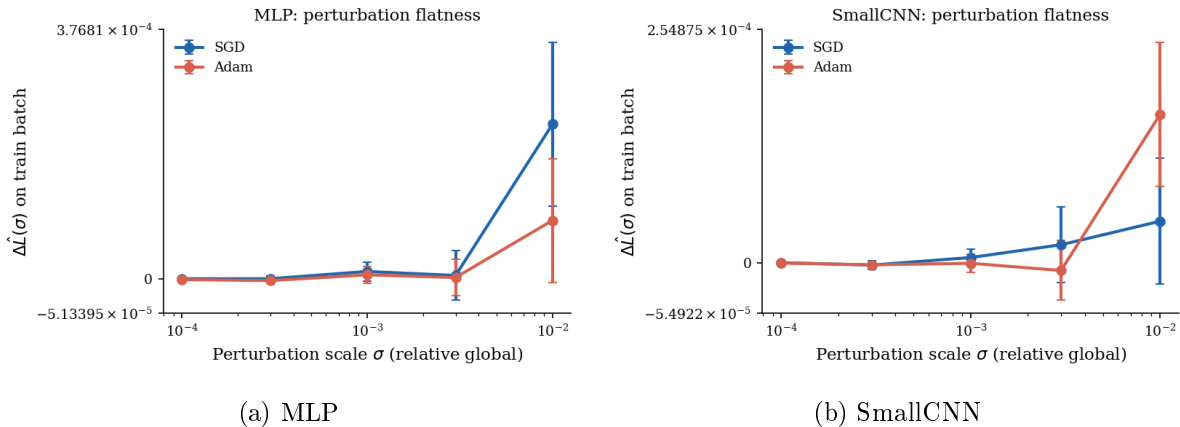


Figure 7: Mean training-loss increase  $\Delta \widehat{L}(\sigma)$  under relative isotropic Gaussian perturbations, averaged over five matched seeds and  $K = 10$  perturbations per  $\sigma$ . Values at  $\sigma \leq 3 \times 10^{-3}$  are at the noise floor ( $\leq 10^{-5}$ ); the operational comparison is at  $\sigma = 10^{-2}$ .



Across  $\sigma \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ ,  $\widehat{\Delta L}$  is at the  $\leq 10^{-5}$  level for all four (architecture, optimizer) configurations and is statistically indistinguishable from zero given the seed variability; these data points primarily establish a noise floor. We therefore restrict the operational comparison to the largest probed magnitude,  $\sigma = 10^{-2}$ . On the MLP,  $\widehat{\Delta L} = (2.3 \pm 1.2) \times 10^{-4}$  for SGD versus  $(0.9 \pm 0.9) \times 10^{-4}$  for Adam, a  $\sim 2.5\times$  gap in the direction predicted by Part II’s trace ratio; the seed standard deviations are well separated from zero for both optimizers. On the SmallCNN, the values are  $(0.5 \pm 0.7) \times 10^{-4}$  for SGD and  $(1.6 \pm 0.8) \times 10^{-4}$  for Adam: the central tendency is in the same direction as the loss-matched  $\lambda_{\max}$  reversal of Section 9, but both seed standard deviations overlap zero, so this is a directional observation from  $N = 5$  seeds rather than a statistically significant reversal in its own right. We do not interpret the small- $\sigma$  data points as a separate test of the quadratic prediction  $\widehat{\Delta L} \propto \sigma^2 \text{tr}(H)$ ; the perturbation evidence corroborates the Hessian readings only at the operational  $\sigma = 10^{-2}$  contrast, and only on the MLP at one-sigma confidence.

## 11 Part VII: Linear Mode Connectivity

### 11.1 Motivation

Part II reports that Adam’s solutions are far from SGD’s in Euclidean parameter space. A natural follow-up question is whether the two solutions sit in the same low-loss basin or in different basins separated by a loss barrier. Linear mode connectivity [6, 7] probes this directly: if a straight-line interpolant between two minima never rises above the larger of their two losses, the minima lie in a connected low-loss region.

### 11.2 Method

For each matched seed and architecture, define the line

$$\theta(\lambda) = (1 - \lambda) \theta_{\text{SGD}} + \lambda \theta_{\text{Adam}}, \quad \lambda \in \{0, 0.1, 0.2, \dots, 1.0\}.$$

We evaluate training and test loss at each  $\lambda$ . For BatchNorm models, the running statistics interpolate to values that no longer match the data, so we recalibrate them at each interior  $\lambda$  by a single forward pass over 50 training batches in train mode (no parameter update). The endpoints  $\lambda = 0$  and  $\lambda = 1$  keep their original BN stats. The loss *barrier* is

$$B = \max_{\lambda \in [0, 1]} \widehat{L}(\theta(\lambda)) - \max\{\widehat{L}(\theta_{\text{SGD}}), \widehat{L}(\theta_{\text{Adam}})\}.$$

$B \approx 0$  indicates linear connectivity;  $B > 0$  indicates a barrier.

### 11.3 Results

Figure 8 plots  $\widehat{L}(\theta(\lambda))$  over the interpolation, and Table 8 reports  $B$  averaged over seeds.

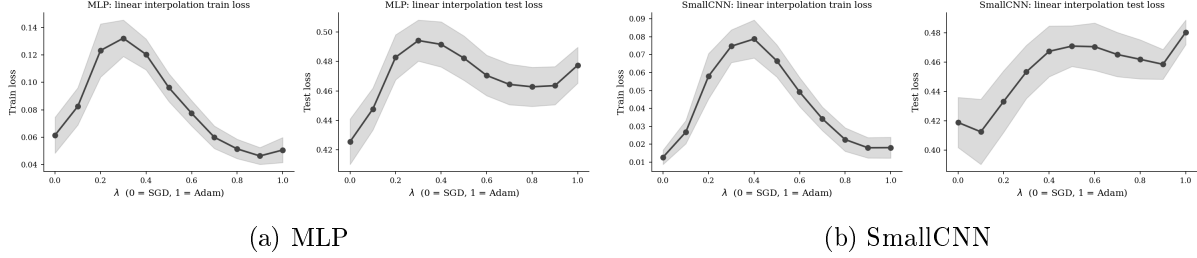


Figure 8: Linear mode connectivity: training and test loss along  $\theta(\lambda) = (1 - \lambda)\theta_{\text{SGD}} + \lambda\theta_{\text{Adam}}$ , with BatchNorm recalibrated at interior  $\lambda$ . Solid line is mean over five matched seeds; band is  $\pm 1$  std.

| Model    | Comparison                 | Train barrier $B_{\text{train}}$ | Test barrier $B_{\text{test}}$ |
|----------|----------------------------|----------------------------------|--------------------------------|
| MLP      | SGD-vs-Adam (matched seed) | $0.0721 \pm 0.0086$              | $0.0174 \pm 0.0086$            |
| MLP      | SGD-vs-SGD (cross seed)    | $0.4952 \pm 0.0630$              | $0.2855 \pm 0.0614$            |
| MLP      | Adam-vs-Adam (cross seed)  | $0.4925 \pm 0.0401$              | $0.2528 \pm 0.0389$            |
| SmallCNN | SGD-vs-Adam (matched seed) | $0.0615 \pm 0.0078$              | $0.0032 \pm 0.0065$            |
| SmallCNN | SGD-vs-SGD (cross seed)    | $1.1843 \pm 0.2540$              | $0.8102 \pm 0.2547$            |
| SmallCNN | Adam-vs-Adam (cross seed)  | $1.4214 \pm 0.2918$              | $0.9975 \pm 0.2918$            |

Table 8: Loss barrier along the linear interpolation between two trained solutions. Top sub-block per architecture: matched-seed SGD-vs-Adam (mean  $\pm$  std over five seeds). Lower sub-blocks: same-optimizer cross-seed baseline averaged over  $C(5, 2) = 10$  seed pairs. The matched-seed cross-optimizer barrier is roughly an order of magnitude smaller than either same-optimizer cross-seed baseline.

The matched-seed cross-optimizer training-loss barriers are small but consistently positive (0.0721 for the MLP, 0.0615 for the SmallCNN), and substantially larger than the cross-seed standard deviations: there is a real, reproducible bump in training loss along the straight line connecting matched SGD and Adam solutions, even after BatchNorm recalibration. The same-optimizer cross-seed baselines, however, give a far higher barrier scale:  $B_{\text{train}}^{\text{SS}} = 0.4952 \pm 0.0630$  and  $B_{\text{train}}^{\text{AA}} = 0.4925 \pm 0.0401$  on the MLP, and  $1.1843 \pm 0.2540$  /  $1.4214 \pm 0.2918$  on the SmallCNN. The cross-optimizer matched-seed barrier is therefore roughly  $7\times$  smaller than the same-optimizer cross-seed barrier on the MLP and  $\sim 20\times$  smaller on the SmallCNN. The basin separation between SGD and Adam at matched seed is real but, in absolute terms, far smaller than the basin separation between two same-optimizer runs from different initializations. The large same-optimizer cross-seed barrier is consistent with the instability finding of Frankle et al. [7]: in standard training, two SGD runs that differ only in initialization land in distinct linearly disconnected basins, and our SGD-vs-SGD and Adam-vs-Adam baselines reproduce this effect at our scale. What drives basin connectivity in this protocol is therefore primarily the shared seed (initialization plus minibatch order), not optimizer agreement; the cross-optimizer effect is a small additional barrier on top of an otherwise highly connected matched-seed landscape. Test-loss barriers tell the same story at lower magnitude:  $0.0174$  /  $0.0032$  for matched-seed cross-optimizer pairs versus  $\sim 0.25$ – $1.0$  for same-optimizer cross-seed pairs.

## 12 Part VIII: Function-Space Similarity

### 12.1 Motivation

Parameter-space distance does not directly measure how different the trained networks are as functions. Two networks can have very different parameter vectors and still produce nearly identical predictions on the data distribution, especially in the presence of scale symmetries and Batch-Norm. To check whether the optimizer-induced parameter differences carry through to functional differences, we evaluate three function-level metrics on the test set.

### 12.2 Metrics

For matched SGD and Adam models with predicted class probabilities  $p_S(\cdot|x)$  and  $p_A(\cdot|x)$  and pre-softmax logits  $z_S(x), z_A(x)$ , we compute, over the  $N$ -point test set:

$$D_{\text{pred}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\arg \max_k p_S(k|x_i) \neq \arg \max_k p_A(k|x_i)], \quad (1)$$

$$D_{\text{SKL}} = \frac{1}{2N} \sum_{i=1}^N [D_{\text{KL}}(p_S \| p_A) + D_{\text{KL}}(p_A \| p_S)], \quad (2)$$

$$C_{\text{logit}} = \frac{1}{N} \sum_{i=1}^N \frac{z_S(x_i)^\top z_A(x_i)}{\|z_S(x_i)\|_2 \|z_A(x_i)\|_2}. \quad (3)$$

$D_{\text{pred}}$  is a strict decision-level disagreement;  $D_{\text{SKL}}$  measures the divergence of the full softmax distribution;  $C_{\text{logit}}$  is scale-invariant and captures direction similarity in logit space.

### 12.3 Results

Table 9 reports the three metrics.

| Model    | Comparison                 | $D_{\text{pred}}$   | $D_{\text{SKL}}$    | $C_{\text{logit}}$  |
|----------|----------------------------|---------------------|---------------------|---------------------|
| MLP      | SGD-vs-Adam (matched seed) | $0.0738 \pm 0.0028$ | $0.1749 \pm 0.0051$ | $0.6122 \pm 0.0058$ |
| MLP      | SGD-vs-SGD (cross seed)    | $0.0868 \pm 0.0067$ | $0.2056 \pm 0.0286$ | $0.9187 \pm 0.0035$ |
| MLP      | Adam-vs-Adam (cross seed)  | $0.0822 \pm 0.0042$ | $0.2348 \pm 0.0195$ | $0.9561 \pm 0.0022$ |
| SmallCNN | SGD-vs-Adam (matched seed) | $0.0671 \pm 0.0042$ | $0.2676 \pm 0.0219$ | $0.6516 \pm 0.0061$ |
| SmallCNN | SGD-vs-SGD (cross seed)    | $0.0740 \pm 0.0045$ | $0.3121 \pm 0.0324$ | $0.9050 \pm 0.0039$ |
| SmallCNN | Adam-vs-Adam (cross seed)  | $0.0786 \pm 0.0031$ | $0.4038 \pm 0.0212$ | $0.9295 \pm 0.0040$ |

Table 9: Function-space similarity on the Fashion-MNIST test set. Top sub-block per architecture: matched-seed SGD-vs-Adam (mean  $\pm$  std over five seeds). Lower sub-blocks: same-optimizer cross-seed baseline averaged over  $C(5, 2) = 10$  pairs.  $D_{\text{pred}}$  and  $D_{\text{SKL}}$  are at or below the same-optimizer cross-seed baseline; only the logit-direction cosine  $C_{\text{logit}}$  shows a sharp matched-seed cross-optimizer signal.

The two matched-seed predictors disagree on 7.4% of MLP test predictions and 6.7% of SmallCNN predictions. The same-optimizer cross-seed baselines on  $D_{\text{pred}}$  are  $0.0868 \pm 0.0067$  (SGD-vs-SGD) and  $0.0822 \pm 0.0042$  (Adam-vs-Adam) on the MLP, and  $0.0740 \pm 0.0045$  /  $0.0786 \pm 0.0031$  on the SmallCNN. The matched-seed cross-optimizer  $D_{\text{pred}}$  is therefore at or slightly *below* the same-optimizer cross-seed baseline; symmetric KL behaves the same way (0.1749 for cross-optimizer

matched seed on the MLP, versus 0.20–0.23 for same-optimizer cross-seed). Decision-level and softmax-level disagreement at this protocol scale is not driven by the choice of optimizer above the level of seed-induced disagreement. The exception is logit-direction cosine:  $C_{\text{logit}} \approx 0.61\text{--}0.65$  between matched SGD and Adam, versus 0.92–0.96 within either optimizer across seeds. Optimizer choice does therefore shift the pre-softmax logit *direction* substantially, but argmax decisions and full softmax distributions absorb most of that shift. The architecture comparison adds a secondary effect: the SmallCNN has slightly lower  $D_{\text{pred}}$  but higher  $D_{\text{SKL}}$  than the MLP, indicating that when the convolutional models do disagree they tend to disagree more confidently in distribution.

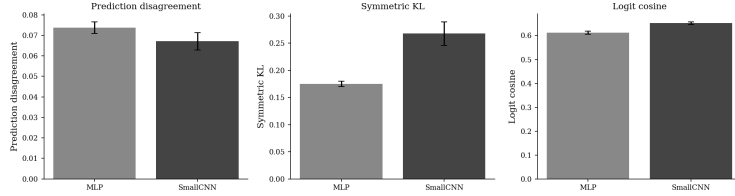


Figure 9: Function-space disagreement metrics on Fashion-MNIST test set, mean  $\pm$  std over five seeds.

## 13 Part IX: Representation-Space Geometry

Function-space metrics compare outputs, but two models can have similar outputs while using different internal representations. We therefore compare penultimate-layer features using linear centered kernel alignment (CKA) [8]. Let  $F_S$  and  $F_A$  denote centered feature matrices for matched SGD and Adam models on the same test subset. Linear CKA is

$$\text{CKA}(F_S, F_A) = \frac{\|F_S^\top F_A\|_F^2}{\|F_S^\top F_S\|_F \|F_A^\top F_A\|_F}.$$

It is insensitive to isotropic rescaling and is a more stable representation-similarity diagnostic than raw feature distance.

| Model    | Comparison                 | Feature kernel alignment A | Linear CKA          |
|----------|----------------------------|----------------------------|---------------------|
| MLP      | SGD-vs-Adam (matched seed) | $0.9697 \pm 0.0010$        | $0.8578 \pm 0.0059$ |
| MLP      | SGD-vs-SGD (cross seed)    | $0.9698 \pm 0.0008$        | $0.8613 \pm 0.0038$ |
| MLP      | Adam-vs-Adam (cross seed)  | $0.9714 \pm 0.0008$        | $0.8497 \pm 0.0071$ |
| SmallCNN | SGD-vs-Adam (matched seed) | $0.9664 \pm 0.0027$        | $0.9147 \pm 0.0113$ |
| SmallCNN | SGD-vs-SGD (cross seed)    | $0.9774 \pm 0.0006$        | $0.9220 \pm 0.0033$ |
| SmallCNN | Adam-vs-Adam (cross seed)  | $0.9677 \pm 0.0024$        | $0.8920 \pm 0.0107$ |

Table 10: Representation-space alignment between penultimate-layer features. Matched cross-optimizer values are close to same-optimizer cross-seed baselines, suggesting that optimizer choice does not strongly reshape representation geometry at this protocol scale.

The key result is that representation similarity remains high. Matched SGD-vs-Adam CKA is 0.858 for the MLP and 0.915 for the SmallCNN, close to the corresponding same-optimizer cross-seed baselines. Thus the large parameter-space separation does not become a large representation-space separation. Combined with the function-space results, this suggests a layered effect: optimizer choice strongly changes parameter trajectories, moderately changes logits, but only weakly changes penultimate feature geometry.

## 14 Three-Level Synthesis

The Parts II–IX sequence organizes evidence at three levels of geometry. Table 11 summarizes the picture.

| Level                                | Quantities                                                                                                                                       | What it answers                                                                                      | Verdict (this study)                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter geometry                   | $\ \theta_T - \theta_0\ _2$ , $\mathcal{P}_T$ , $R_T$ , $\lambda_{\max}(H)$ , $\text{tr}(H)$ , $\bar{d}_{\text{seed}}$ , $\Delta\hat{L}(\sigma)$ | Where does the optimizer go, and what is the local curvature?                                        | Adam path length $\sim 3\times$ SGD's in both architectures; directness ratio essentially equal. MLP: Adam $\sim 5\text{--}6\times$ flatter at matched loss. Small-CNN: ordering may reverse, but this is a single-architecture-pair observation with high $\lambda_{\max}$ estimator variance.                                                                      |
| Basin geometry                       | Linear-interpolation loss curve, barrier $B$                                                                                                     | Are the two solutions in the same low-loss region?                                                   | Cross-optimizer matched-seed train barrier (0.072 MLP, 0.062 CNN) is $\sim 7\times\text{--}20\times$ smaller than the same-optimizer cross-seed baseline (0.49 MLP, 1.18–1.42 CNN). The matched-seed protocol drives connectivity; cross-optimizer adds a small additional barrier. Test barrier $\leq 0.02$ for matched-seed pairs.                                 |
| Function and representation geometry | $D_{\text{pred}}$ , $D_{\text{SKL}}$ , $A(K_S, K_A)$ , CKA                                                                                       | $C_{\text{logit}}$ , Do parameter differences correspond to different predictors or representations? | $D_{\text{pred}}$ ( $\sim 7\%$ ) and CKA (0.86–0.91) are within the same-optimizer cross-seed baseline range; only $C_{\text{logit}} \approx 0.6$ (vs. 0.92–0.96 within-optimizer) shows a sharp matched-seed cross-optimizer signal. Optimizer choice reshapes logit direction but not argmax decisions or representation geometry beyond seed noise at this scale. |

Table 11: Three-level synthesis of the implicit-bias evidence.

The data, with the same-optimizer cross-seed baseline added, support a more conservative version of the message we anticipated:

*Adam substantially changes parameter-space trajectory and local Euclidean geometry on the MLP, and the change survives loss-matching control. As the comparison propagates from parameters to predictions, however, much of what looked like a cross-optimizer effect compresses to within the same-optimizer cross-seed baseline:  $D_{\text{pred}}$ ,  $D_{\text{SKL}}$ ,  $A$ , and linear CKA on matched-seed cross-optimizer pairs lie within the range observed between two runs of the same optimizer at different seeds. The matched-seed protocol drives basin connectivity an order of magnitude beyond what shared-optimizer cross-seed pairs achieve; the cross-optimizer matched-seed train barrier is small but nonzero, and the only function-level metric that shows a sharp cross-optimizer signal above seed noise is the logit-direction cosine  $C_{\text{logit}}$ . Architecture mediates which curvature regime each optimizer selects, but the SmallCNN flatness reversal is a single-architecture-pair observation with high  $\lambda_{\text{max}}$  estimator variance and should be replicated.*

This is the core message of the extended report. Optimizer-induced parameter geometry is real, especially on the MLP, and not merely a final-loss artifact; but with a same-optimizer cross-seed baseline in view, the function- and representation-level differences between matched SGD and Adam at this protocol scale are mostly accounted for by seed-induced variability rather than by optimizer-induced bias, with  $C_{\text{logit}}$  as the principal exception.

## 15 Overall Discussion

The main question was whether adaptive optimization alters implicit bias and generalization. The answer is layered. Adam clearly changes parameter-space trajectory: it moves farther from initialization, travels a longer path, and produces more dispersed endpoints. On the MLP, this difference remains visible after loss matching and appears in much lower Hessian curvature and smaller perturbation loss increase. On the CNN, the curvature ordering is less stable, which suggests that architecture can dominate or redirect optimizer-induced geometry.

The effect becomes weaker as we move away from parameters. Linear mode connectivity shows that matched SGD and Adam endpoints are not simply identical points in the same basin, but their train-loss barrier is much smaller than same-optimizer cross-seed barriers. Function-space metrics show nontrivial logit-direction differences, yet prediction disagreement and symmetric KL remain close to same-optimizer cross-seed baselines. Representation CKA is even more compressed: penultimate features remain highly aligned across optimizers. The most concise summary is therefore:

*Adam changes geometry strongly in parameter space, mildly in basin and logit geometry, and weakly in representation geometry.*

This explains why the generalization story is not binary. The MLP shows the strongest flatness advantage for Adam, but Adam does not obtain a clear test-accuracy or gap advantage. The CNN shows similar generalization even when parameter trajectories differ. The conclusion direction is consistent with Wilson et al. [5], who report that adaptive methods can generalize worse than well-tuned SGD; the underlying mechanism in our MLP setting is, however, the opposite of the one they invoke: they attribute the gap to Adam selecting *sharper* minima, whereas in our matched-seed Fashion-MNIST MLP Adam selects substantially *flatter* solutions and still does not generalize better. Thus optimizer-induced implicit bias is real, but its effect on test performance is filtered through architecture, loss level, basin connectivity, functional equivalence, and representation similarity.

## 15.1 Scale-Aware Sharpness and Reparameterization Caveat

The Hessian quantities in this report are Euclidean parameter-space diagnostics. They are useful for controlled within-architecture comparisons, but not invariant measures of function complexity. For a two-layer ReLU network,

$$f(x) = W_2\sigma(W_1x),$$

positive homogeneity gives

$$W_2\sigma(W_1x) = (c^{-1}W_2)\sigma(cW_1x), \quad c > 0.$$

The two parameterizations represent the same function but can have different norms and Hessian spectra. BatchNorm introduces related scale freedoms. This reparameterization caveat [4] is why we interpret  $\lambda_{\max}(H)$ ,  $\text{tr}(H)$ , and perturbation flatness as within-protocol diagnostics rather than universal generalization measures.

## 15.2 Limitations

The study uses one dataset, two small architectures, five seeds, and finite hyperparameter tuning. Hessian estimates are mini-batch proxies, and BatchNorm complicates both sharpness and mode-connectivity interpretation. The no-regularization protocol is useful for isolating optimizer effects, but it is not a production training recipe. The architecture-dependent flatness reversal should therefore be interpreted as a directional observation worth replicating, not as a universal Adam-vs-SGD law.

## 16 Conclusion

This project compared SGD with momentum and Adam as two preconditioned stochastic dynamics. The central mathematical distinction is that SGD uses scalar spatial preconditioning, while Adam uses a time-varying diagonal preconditioner, leading locally to different error dynamics through

$$e_{t+1} \approx (I - P_t H) e_t.$$

Empirically, Adam converges faster than the default SGD baseline and selects different parameter-space regions under matched seeds. These differences are not fully explained by final training loss or by a simple flat-minima story. Loss-matched geometry, perturbation flatness, mode connectivity, function-space similarity, and representation CKA show that the optimizer signal is strongest in parameter space and becomes progressively weaker at the predictor and representation levels. The final lesson is not that Adam or SGD is universally better, but that adaptive preconditioning changes solution selection while generalization depends on how that parameter-space change survives through architecture, basin structure, function space, and representation geometry.

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